

UCE837: SUSTAINABLE & SMART MATERIALS

L	T	P	Cr
3	1	0	3.5

Course objective: To expose the students to the latest trends in the development of sustainable and smart materials in the field of construction industry based on the efficient uses of resources.

Sustainable Materials

Introduction to sustainability concepts and life cycle analysis

Role of Materials: Carbon from Cement, alternative cements and cementitious material, Alternative fuel for cements for reduction in carbon emission, Sustainability issues for concrete

Use of sustainable construction materials- Strength, durability and environmental impact, Material Characterization,

Concrete made with industrial by-products as replacement to cement, coarse or fine aggregates, Concrete with natural sustainable materials

Overview of Life cycle analysis – Life Cycle Assessment of Cement and Concrete, Life Cycle Stages of Buildings, LCA Software tools, Embodied carbon calculation of concrete mix and building

Smart Materials

Definition of smart materials and structures

Different smart materials, properties and applications

Energy Harvesting Materials, energy conservation in built environment using smart materials,

Zero energy buildings

Learning from nature: Bio-inspired materials

Exercises/Assignments:

Case studies related to the application of traditional and advanced technology for smart and sustainable construction.

Course Learning Outcomes (CLO):

Upon completion of this course, the students will be able to:

1. Introduction to concepts of Sustainability and role of construction materials.
2. Develop concrete using various industrial by-products and natural materials in achieving sustainable construction.
3. Use LCA approach for designing a sustainable concrete mix.
4. Use smart materials and smart systems in construction

Text Books

- Brian Culshaw, Smart Structures and Materials, Artech House, 2000
- Suresh Bhalla, 'Piezoelectric materials: Applications in SHM, energy harvesting and biomechanics', Ane Books Pvt Ltd., 2016
- Siddique R, Cachim P. Waste and Supplementary Cementitious Materials in Concrete: Characterisation, Properties and Applications. Woodhead Publishing

Reference Books:

- Baoguo Han, Xun Yu, JinpingOu, Self-sensing Concrete in Smart Structure, Butterworth-

Heinemann and Elsevier, 2014

- Michael Thomas, Supplementary Cementing Material in Concrete, CRC Press, 2013
- Satish Chandra, Waste materials used in concrete, Noyes Publications, 1997

Evaluation Scheme:

Sr.No.	EvaluationElements	Weightage(%)
1.	MST	30
2.	EST	45
3.	Sessional (May includeAssignments/Projects/Tutorials/Quizzes)	25